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ORIGINAL PAPER

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Empirical and projected economic burden of chronic myeloid leukaemia in Sweden from 2015 to 2030: A population-based study

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Summary

Despite therapeutic advances and improved survival, the long-term economic burden of chronic myeloid leukaemia (CML) remains under-recognised given evolving treatment practices and fluctuating drug costs. We aimed to estimate and project total prevalence costs for CML in Sweden from a healthcare sector perspective, representing the direct healthcare expenditures for all patients living with CML, based on real-world drug and procedure prices. We used data from the Swedish Cancer Register and the Swedish CML register to estimate and project prevalence and costs. The estimated numbers of prevalent cases were 1808 (95% confidence interval [CI], 1604–2011) in 2025 and 2120 (95% CI, 1916–2325) in 2030, driven by stable incidence and improved survival. Despite increasing prevalence, the annual total direct healthcare costs for all prevalent CML patients in Sweden decreased from USD 40.04 million (95% CI, 33.70–46.40) in 2015 to USD 26.04 million (95% CI, 23.13–28.97) in 2025, then projected to slightly increase to USD 30.67 million (95% CI, 27.70–33.64) in 2030. While CML prevalence proportions are expected to increase, declining treatment costs may mitigate the burden on the Swedish healthcare system. These population-based projections can inform long-term planning and pricing strategies for CML care.

KEYWORDS
Chronic myeloid, cost of illness, economic burden, leukaemia, prevalence, Sweden

INTRODUCTION

The introduction of tyrosine kinase inhibitors (TKIs) has transformed the treatment of chronic myeloid leukaemia (CML), turning a once fatal disease into a mostly manageable chronic condition. Survival among CML patients has improved substantially and now approaches that of the general population[1]; meanwhile, the incidence of CML has remained relatively stable over recent decades.[2,3] The combination of improved survival and a steady incidence has led to a growing prevalence in Sweden, from 11.9 cases per 100 000 individuals in 2012 to a predicted 15.1 per 100 000 in 2020.[2]

The economic burden of CML in Sweden is evolving, mainly due to declining drug prices linked to novel generic TKIs and emerging new therapies. Initially, TKIs were associated with high costs (more than € 30 000 per patient per year in Sweden[4]), but those costs have been considerably reduced due to patent expiration. Generic imatinib emerged in the Swedish market in December 2016,[5] followed by the first generic dasatinib, which entered the market during June 2020.[6] Further price reductions are expected for nilotinib and bosutinib, corresponding to their patent expirations in Sweden in 2025 and in the European Union (EU) in 2028[7], respectively. A novel third-generation TKI, asciminib has recently been approved in Sweden[8] in a third-line setting with promising molecular response rates and encouraging tolerability, but at higher costs.

Treatment-free remission (TFR) has become an important goal for CML management, based on growing evidence that

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many patients who achieve a lasting deep molecular response (DMR) can safely discontinue therapy.[9–14] Clinical trials such as European Stop Tyrosine Kinase Inhibitor Trial (EURO-SKI)[9] and population-based studies in Sweden[13] suggest that 40%–50% of eligible patients with DMR can enter TFR, although nearly half may have molecular relapse after TFR within 24 months and thus need to re-initiate TKI treatment. Nevertheless, the possibility of temporary or sustained TFR offers a new clinical management with potential for cost savings. For example, EURO-SKI estimated € 22 million in drug cost savings among the study population of 586 patients.[9]

The interplay of incidence, survival, prevalence, drug prices and evolving treatment strategies complicates projections of the future economic burden of CML. A few studies[15–17] have examined the long-term healthcare costs associated with CML. However, none has used nationwide, whole-population cancer registers that fully account for temporal changes in treatment practices, drug pricing and changes in patient composition across different disease states. This study aims to address this gap by estimating the total prevalence costs of CML in Sweden from 2015 to 2030, using data from the Swedish Cancer Register and the Swedish CML register.

METHODS

Study design and data sources

We used several nationwide Swedish registers, including the Swedish Cancer Register and the Swedish CML register, to obtain data on survival, demographics, disease characteristics and treatment patterns for patients diagnosed with CML between ages 18 and 99 years. For cost data, we used individual-level data from the Swedish CML register and the National Prescribed Drug Register, as well as aggregate-level data from several official price lists and studies. Table 1 provides a summary of the data sources used in the analysis. Detailed definitions of variables and data extraction procedures are presented in Section S1.

Estimating and projecting prevalence and total prevalence costs

We used the Prevalence, Incidence, Analysis MODel method[23] to estimate and project prevalence, incorporating incidence rates and net survival (described below) together with the population and mortality data of Sweden.

We estimated total prevalence costs from a healthcare sector perspective.[24] Total prevalence costs refer to the total healthcare expenditures incurred for all individuals living with CML. For each year, the costs were obtained by multiplying the average yearly cost per patient in each disease state by the number of patients in that state and summing across all states. Full statistical details and the overview workflow are provided in Section S2.

Estimating and projecting of incidence

Because prevalence estimates depend on both incidence and survival, we first modelled CML incidence using an age–period–cohort (APC) model fitted to observed data from the Swedish Cancer Register between 1973 and 2019. The final model was then used to project CML incidence rates from

TABLE 1 Overview of data sources.

Data type	Data source	Period
Survival (CML patients)	Swedish Cancer Register, Cause of Death Register	Diagnosed 1973–2019 with follow-up until 2020
Mortality (general population)	Human Mortality Database[18]	Up to 2023 (projected constant thereafter)
Population size (general population)	Human Mortality Database[18]	Up to 2024 (projected constant thereafter)
Disease phase at diagnosis (CML patients)	Swedish CML register	Diagnosed 2002–2023
TKI treatment patterns	Swedish CML register, National Prescribed Drug Register	Diagnosed 2007–2017 with follow-up until 2018
TFR	Swedish CML register	Diagnosed 2007–2012 with follow-up until 2018
Disease progression to AP/BP	Swedish CML Register, Swedish Cancer Register	Diagnosed 2007–2017 with follow-up until 2018
AlloSCT procedures	Swedish CML register, Swedish Patient Register	Diagnosed 2007–2017 with follow-up until 2018
Costs for TKI (individual-level)	Swedish CML register, National Prescribed Drug Register	Diagnosed 2007–2017 with follow-up until 2018
Costs for TKI (aggregate-level)	TLV Price and Decision Database[19]	2019–2025 (projected constant thereafter)
Costs for alloSCT and progression	Hernlund et al.,[20] adjusted yearly by the Extra-regional Pricing List for EU/EEA and Switzerland[21]	Diagnosed 2007–2015
Costs for laboratory testing	National Price List for Laboratory Medical Services[22]	2024

Note: Details of variable definitions of each data type are provided in Section S1.

Abbreviations: AlloSCT, allogeneic stem cell transplantation; AP, accelerated phase; BP, blastic phase; CML, chronic myeloid leukaemia; TFR, treatment-free remission; TKI, tyrosine kinase inhibitor; TLV, the Dental and Pharmaceutical Benefits Agency (Tandvårds- och läkemedelsförmånsverket).

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2020 to 2030. Details of the APC modelling approach are provided in Section S3.

Estimating and projecting of net survival

Net survival was estimated from individual-level data on CML patients diagnosed between 1973 and 2019, representing the probability of surviving CML in the absence of other causes of death.[25] To reflect different future survival assumptions, we considered two scenarios for projecting prevalence, both within a relative survival framework[26]:

- Scenario 1—Fixed survival (conservative projection): We estimated period-specific in 3-year intervals net survival for patients diagnosed during 1973–1975, 1975–1977, …, and 2015–2019 using the Pohar–Perme estimator.[27] We assumed no improvements in net survival beyond 2019, fixed at the 2019 level for projections from 2020 to 2030.
- Scenario 2—Dynamic survival (optimistic projection): We fitted a flexible parametric relative survival model[28] to the observed data up to 2019 and projected survival from 2020 to 2030 using the estimated model coefficients. This approach allowed for continued improvements over time by extrapolating the temporal trends.

Scenario 1 served as the base case for subsequent analyses, with Scenario 2 results reported in sensitivity analyses. Full statistical methods for both scenarios are presented in Section S4.

Estimating the distribution of prevalent CML patients across disease states

Patients with CML can be classified into various disease states.[29] To estimate the distribution of patients across these states over time, we used both observed (2000–2019) and projected (2020–2030) incidence data. We estimated the distribution of patients across seven disease states over time—including first-line TKI (1L TKI), second-line TKI (2L TKI), third-line or later TKI (3L+ TKI), TFR, progression, allogeneic stem cell transplantation (alloSCT) and death using the CML natural history model previously developed by Chen et al.[29] Methodological details for this model are described elsewhere.[29]

Finally, we multiplied the estimated proportion of each state by the total number of prevalent cases to determine the number of patients in each disease state from 2015 to 2030. Details of the estimation procedures are provided in Section S5 and Table S1.

Estimating average yearly cost per patient

TKI costs were estimated using data from the Swedish CML register, the National Prescribed Drug Register and the Price and Decision Database from the Dental and Pharmaceutical Benefits Agency (Tandvårds- och läkemedelsförmånsverket, TLV). Costs for progression, alloSCT and other health resources were informed by the published literature[4,20] and Sweden's official price lists.[21,22] All costs were inflated to 2024 Swedish krona (SEK) using the consumer price index from Statistics Sweden[30] and converted to USD using the 2024 exchange rate (1 USD = 10.56 SEK) from Sveriges Riksbank (the central bank of Sweden).[31] Detailed cost calculations for the average yearly cost per patient are provided in Section S6 and Tables S2–S4.

We did not apply discounting to future costs, as our study aims to examine trends in economic burden rather than evaluate the cost-effectiveness of specific interventions. However, for readers who would like to apply our results to assess specific interventions, we have provided the results with 3% and 5% discounting rates to costs from 2025 onwards in Tables S5 and S6 respectively.

Sensitivity analysis

We performed one-way sensitivity analyses to evaluate the influence of various parameters on the projections. The parameters considered were as follows: (1) Scenario 2, representing dynamic survival; (2) variations in discount rates on costs: 3% and 5% applied from 2025 onwards (Tables S5 and S6), if a decision maker wishes to evaluate any intervention using our projections; (3) a 25% increase in yearly cost of 3L+ TKI in 2024, reflecting the market introduction of asciminib as a third-line TKI in 2023[8]; (4) a 25% decrease in yearly 3L+ TKI costs in 2025, consequent to the potential entry of generic bosutinib in 2024[7]; (5) a ±25% variation in yearly 1L TKI costs in 2029; (6) a ±25% variation in yearly 2L TKI costs in 2029; and (7) a ±25% variation in yearly 3L+ TKI costs in 2029, reflecting potential price reductions following the potential introduction of generic nilotinib and ponatinib[7] respectively.

We developed an interactive R Shiny application (R version 4.4.2) to facilitate sensitivity analyses on prevalent cases, yearly cost per patient and total prevalence costs. The app integrates editable tables, plots and with options to apply discounting rate, allowing users to reproduce and extend our analyses. It is publicly accessible at: https://enochytchen.shinyapps.io/CMLEcoBurdenSE/.

Role of the funding source and ethical approval

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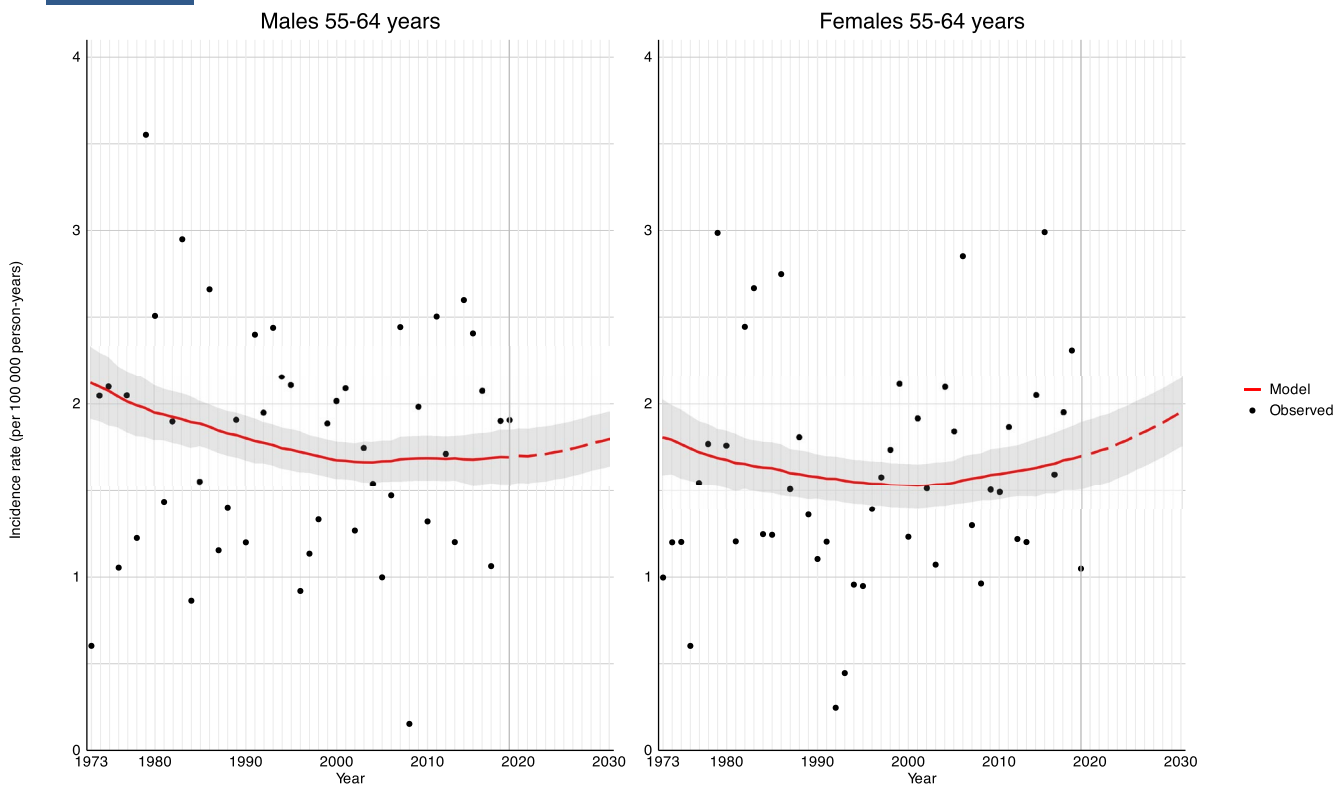


FIGURE 1 Observed and predicted crude incidence rates (per 100 000 person-years) for male and female patients with chronic myeloid leukaemia aged 55–64 years at diagnosis in Sweden during 1973–2030. Black dots represent observed incidence rates. Red lines represent the estimates by the age–period–cohort model, with shaded areas indicate 95% confidence intervals. Dashed lines after 2019 represent projections beyond the observed period.

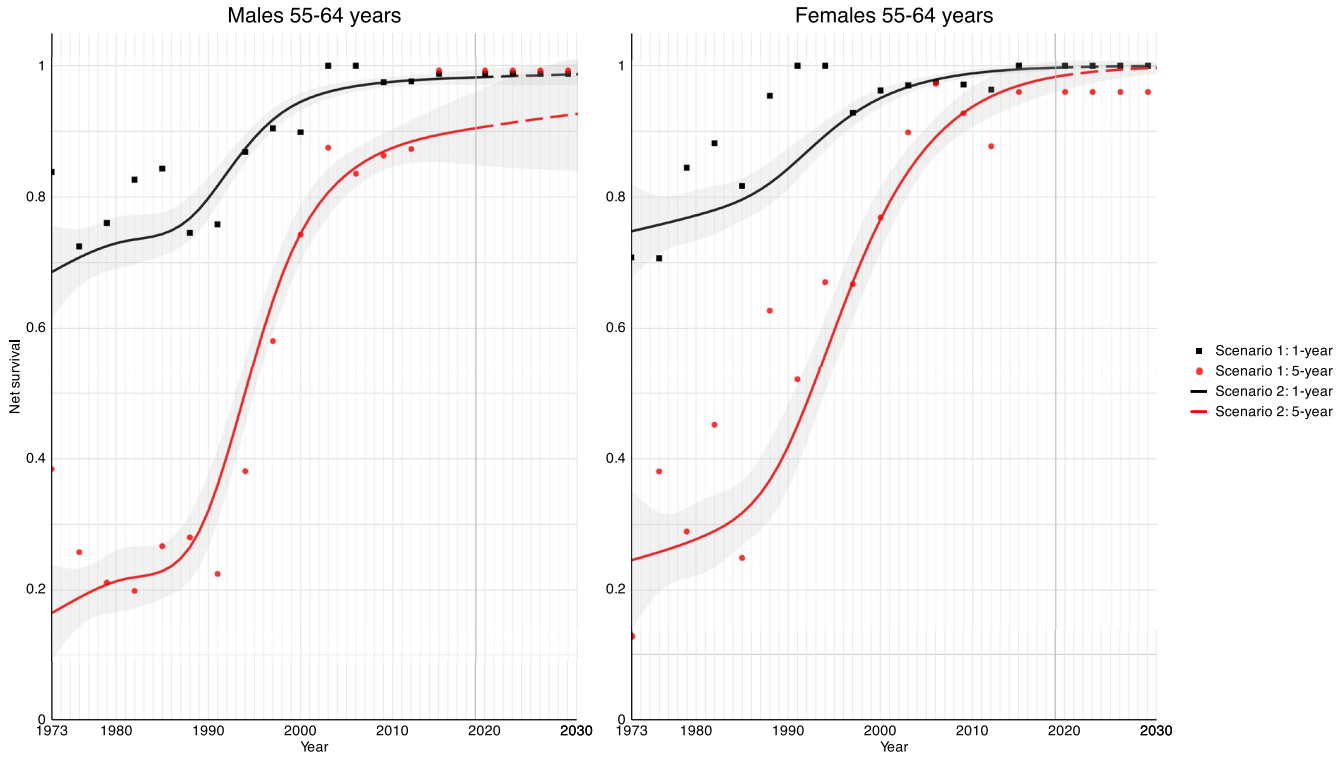


FIGURE 2 Observed and projected net survival for male and female patients with chronic myeloid leukaemia aged 55–64 years at diagnosis in Sweden during 1973–2030. Black and red points represent observed 1- and 5-year net survival estimates by the Pohar–Perme estimator respectively (Scenario 1: fixed survival). For patients diagnosed after 2019, survival was assumed to remain at the level estimated for the 2015–2019 cohort. Black and red lines represent 1- and 5-year net survival estimates by the flexible parametric relative survival model, respectively, with 95% confidence intervals (Scenario 2: dynamic survival).

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RESULTS

The observed and predicted crude incidence rates for CML among patients aged 55–64 years at diagnosis from 1973 to 2030, stratified by sex, are shown in Figure 1. In general, the predicted incidence rates for males and females followed the same trend. Although male incidence was higher than female in the 1970s, our model predicted a higher incidence rate in females, 2.0 per 100 000 person-years (95% confidence intervals [CI]: 1.8–2.2), than in males, 1.8 per 100 000 person-years (95% CI: 1.6–2.0), in 2030. We presented the crude incidence rates of all age groups in Figure S2.

Net survival after a CML diagnosis has dramatically improved over recent decades. Figure 2 displays survival trends for patients aged 55–64 years at diagnosis under two different scenarios, shown here as a representative example. In Scenario 1 (fixed survival), net survival rose steadily from the mid-1970s to early 2000s; by 2015–2017, 1-year net survival exceeded 95% and 5-year survival reached roughly 80%–90%, with similar patterns in both sexes. For patients diagnosed after 2019, survival was assumed to remain at the level estimated for the 2015–2019 cohort. Net survival estimates for all age groups are provided in Figure S3 (Scenario 1: fixed survival) and Figure S4 (Scenario 2: dynamic survival).

Projections of CML incidence and prevalence under Scenario 1 (fixed survival) and Scenario 2 (dynamic survival) are provided in Table 2 and illustrated by prevalence curves in Figure S5. Overall, the estimates in prevalence were similar but slightly higher in Scenario 2 than in Scenario 1. Similarly, the increase in the predicted number of prevalent cases from 2015 to 2030 was greater in Scenario 2 (82.9%, 95% CI: 72.5%–97.2%) than in Scenario 1 (75.9%, 95% CI: 66.5%–88.9%), which reflects the change in prevalence over that period.

From 2015 to 2030, CML in Sweden shows an increasing number of prevalent cases but declining average yearly costs, resulting in reduced total prevalence costs. These trends are illustrated in Figure 3 and summarised in Table 3: total prevalence costs decreased from USD 40.04 million (95% CI 33.70–46.40) in 2015 to USD 26.04 million (95% CI 23.13–28.97) in 2025, and then slightly increased to USD 30.67 million (95% CI 27.70–33.64) in 2030, as the average yearly cost per patient decreased from USD 33 224 in 2015 to USD 14 408 in 2025 and USD 14 464 in 2030. Patient numbers are expected to grow mainly in 1L and 2L TKI (Table 3). First-year alloSCT is expected to be the costliest state (more than USD 150 000 per patient yearly), while the progression cost is at USD 20 000–30 000. TKI costs will decline over time: average 1L costs drop from USD 35 244 in 2015 to USD 8063 in 2025, and 2L/3L+ costs fall from roughly USD 40 000 to USD 20 000. TFR is consistently predicted to be the least expensive state (USD 7558 in 2015 and USD 6030 in 2025).

Results from the one-way sensitivity analyses for total prevalence costs in 2025 and 2030 are presented in Figure 4.

TABLE 2 Population, estimated incidence and prevalence of chronic myeloid leukaemia aged 18–99 years at diagnosis in Sweden under two scenarios: (1) fixed and (2) dynamic survival, in 2015, 2020, 2025 and 2030, presented with crude and age-standardised (to the 2000 Swedish population) estimates with 95% confidence interval.

Scenario	Year	Population in thousands	Incident cases	Incidence per 100 000 person-years	Age-standardised incidence per 100 000 person-years	Prevalent cases	Prevalence per 100 000 persons	Age-standardised prevalence per 100 000 persons
Scenario 1. Fixed survival	2015	9745	103.3 (99.6–107.0)	1.06 (0.98–1.14)	1.02 (0.94–1.10)	1205.4 (1014.3–1396.5)	12.36 (10.40–14.33)	11.84 (9.88–13.80)
	2020	10 255	113.2 (108.4–117.9)	1.11 (1.01–1.20)	1.06 (0.96–1.15)	1497.4 (1296.4–1698.4)	14.60 (12.64–16.56)	13.87 (11.91–15.83)
	2025	10 363	121.9 (117.1–126.7)	1.18 (1.08–1.27)	1.10 (1.00–1.20)	1807.5 (1604.3–2010.6)	17.44 (15.48–19.40)	16.05 (14.09–18.01)
	2030	10 420	130.4 (125.7–135.2)	1.25 (1.15–1.35)	1.16 (1.06–1.26)	2120.3 (1916.1–2324.6)	20.35 (18.39–22.31)	18.24 (16.28–20.20)
Scenario 2. Dynamic survival	2015	9745	103.3 (99.6–107.0)	1.06 (0.98–1.14)	1.02 (0.94–1.10)	1209.3 (1018.2–1400.4)	12.41 (10.45–14.37)	11.92 (9.96–13.89)
	2020	10 255	113.2 (108.4–117.9)	1.11 (1.01–1.20)	1.06 (0.96–1.15)	1543.0 (1342.0–1744.0)	15.05 (13.09–17.01)	14.30 (12.34–16.26)
	2025	10 363	121.9 (117.1–126.7)	1.18 (1.08–1.27)	1.10 (1.00–1.20)	1881.8 (1678.7–2084.9)	18.15 (16.19–20.11)	16.65 (14.69–18.61)
	2030	10 420	130.4 (125.7–135.2)	1.25 (1.15–1.35)	1.16 (1.06–1.26)	2211.9 (2007.7–2416.2)	21.23 (19.27–23.19)	18.90 (16.94–20.86)

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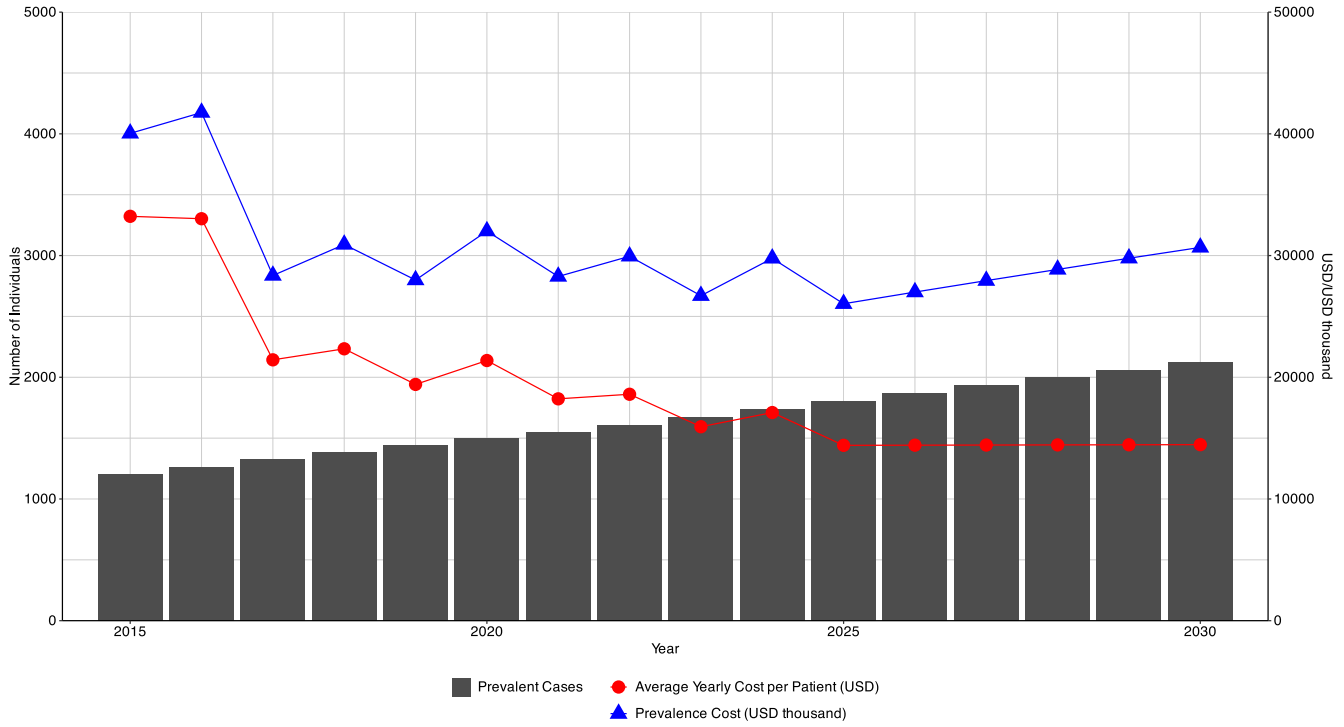


FIGURE 3 Estimated and projected prevalent cases, average yearly cost per patient (USD), total prevalence costs (USD thousand) of chronic myeloid leukaemia in Sweden from 2015 to 2030.

Using dynamic survival (Scenario 2) had only a modest effect on total prevalence costs, with increases of USD 0.43 million in 2025 and USD 0.57 million in 2030. In contrast, changes in 3L+ TKI pricing had a more notable impact: a 25% cost increase in 2024 led to an increase of USD 1.62 million in 2025, while a 25% price decrease in 2025 reduced costs by USD 1.62 million. The most substantial change was observed in 2030, where applying a discounting rate of 5% from 2025 and onwards reduced total prevalence costs by USD 7.78 million. Adjusting 1L, 2L and 3L+ TKI costs by ±25% from 2029 resulted in cost differences of USD ±1.46, ±2.06, and ±2.00 million respectively. An interactive R Shiny app is available to reproduce and further explore the sensitivity analyses.

DISCUSSION

In this study, we projected a steady increase in the number of individuals living with CML in Sweden from 2015 to 2030 under both fixed and dynamic survival scenarios, despite stable incidence. The increase was predominantly driven by the rising numbers of patients on 1L and 2L TKIs. Notably, while the prevalent cases increased over the years, the total prevalence costs decreased from USD 40.04 million (95% CI 33.70–46.40) in 2015 to USD 30.67 million (95% CI 27.70–33.64) in 2030. Despite a growing number of patients, the total prevalence costs remained relatively stable mainly because the average yearly cost of TKI treatment declined over time.

Our projections highlight the uncertainty introduced by our survival assumptions. The fixed survival scenario (Scenario 1) may be overly conservative, as net survival, particularly in younger patients, would likely improve over time (Figure S3). Conversely, the dynamic survival scenario (Scenario 2) may be overly optimistic for older age groups; for instance, it predicted 5- and 10-year net survival exceeding 90% for patients aged 75–99 years after 2020 (Figure S4), a pattern likely reflecting model extrapolation rather than clinical reality.

The declining cost trends observed in our analysis were primarily driven by changes in TKI pricing. The introduction of generic imatinib[5] in December 2016 and generic dasatinib[6] in June 2020 substantially reduced 1L to 3L+ costs in Sweden. Future potential generic entry of nilotinib and bosutinib, following expected EU patent expiry in 2028,[7] may further reduce TKI costs. Assuming these generics reach price levels comparable to today's generic imatinib (approximately 800 USD per patient per year), their yearly cost per patient could decline by an estimated 60%–80% relative to 2025 prices. While TFR contributes to cost savings, its overall impact on total prevalence costs is modest, as approximately half of patients initiating TFR resume treatment within 24 months.[13] Likewise, while first-year alloSCT remains highly expensive, only a small number of patients fall within this state annually,[3] resulting in a limited effect on total prevalence costs. The same applies to patients in progression (accelerated phase/blastic phase), who also represent a small proportion of the CML population.

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TABLE 3 Prevalent cases, yearly cost per patient and prevalence costs by state and overall total.

	1L TKI			2L TKI			3L TKI			Progression		
Year	Prevalent cases	YCPP (USD)	Prevalence costs (USD m)	Prevalent cases	YCPP (USD)	Prevalence costs (USD m)	Prevalent cases	YCPP (USD)	Prevalence costs (USD m)	Prevalent cases	YCPP (USD)	Prevalence costs (USD m)
2015	523.2 (440.3, 606.2)	35 244	18.44 (15.52, 21.36)	284.0 (239.0, 329.0)	40 467	11.49 (9.67, 13.31)	162.3 (136.6, 188.1)	39 520	6.41 (5.40, 7.43)	74.3 (62.5, 86.1)	19 269	1.43 (1.20, 1.66)
2016	542.5 (459.7, 625.4)	34 245	18.58 (15.74, 21.42)	293.7 (248.8, 338.5)	40 264	11.83 (10.02, 13.63)	175.3 (148.5, 202.1)	36 216	6.35 (5.38, 7.32)	78.8 (66.7, 90.8)	23 465	1.85 (1.57, 2.13)
2017	547.2 (466.2, 628.2)	13 311	7.28 (6.21, 8.36)	311.8 (265.7, 358.0)	35 929	11.20 (9.55, 12.86)	192.4 (163.9, 220.9)	28 093	5.41 (4.60, 6.21)	81.7 (69.6, 93.8)	23 557	1.92 (1.64, 2.21)
2018	565.6 (484.5, 646.7)	12 114	6.85 (5.87, 7.83)	323.9 (277.5, 370.3)	36 851	11.94 (10.23, 13.65)	208.7 (178.8, 238.6)	34 393	7.18 (6.15, 8.21)	79.2 (67.9, 90.6)	24 845	1.97 (1.69, 2.25)
2019	564.5 (486.0, 643.0)	9201	5.19 (4.47, 5.92)	338.9 (291.8, 386.0)	31 735	10.75 (9.26, 12.25)	226.8 (195.3, 258.3)	28 623	6.49 (5.59, 7.39)	88.4 (76.1, 100.7)	25 630	2.27 (1.95, 2.58)
2020	579.2 (501.5, 657.0)	12 108	7.01 (6.07, 7.95)	344.4 (298.2, 390.6)	32 246	11.11 (9.62, 12.60)	241.4 (209.0, 273.8)	32 641	7.88 (6.82, 8.94)	94.6 (81.9, 107.2)	26 718	2.53 (2.19, 2.86)
2021	592.3 (515.3, 669.2)	10 727	6.35 (5.53, 7.18)	352.5 (306.7, 398.2)	24 141	8.51 (7.40, 9.61)	255.5 (222.3, 288.7)	27 897	7.13 (6.20, 8.05)	99.7 (86.8, 112.7)	26 628	2.65 (2.31, 3.00)
2022	607.6 (531.4, 683.8)	11 228	6.82 (5.97, 7.68)	362.8 (317.3, 408.4)	23 899	8.67 (7.58, 9.76)	270.4 (236.5, 304.4)	28 010	7.57 (6.62, 8.53)	103.3 (90.3, 116.2)	25 229	2.61 (2.28, 2.93)
2023	627.2 (551.4, 703.0)	9147	5.74 (5.04, 6.43)	375.2 (329.9, 420.6)	19 665	7.38 (6.49, 8.27)	286.4 (251.8, 321.0)	23 365	6.69 (5.88, 7.50)	104.7 (92.1, 117.4)	25 264	2.65 (2.33, 2.97)
2024	641.6 (566.9, 716.3)	10 943	7.02 (6.20, 7.84)	388.7 (343.5, 434.0)	19 847	7.71 (6.82, 8.61)	302.6 (267.4, 337.9)	25 048	7.58 (6.70, 8.46)	111.6 (98.6, 124.6)	26 409	2.95 (2.60, 3.29)
2025	656.4 (582.7, 730.2)	8063	5.29 (4.70, 5.89)	401.4 (356.3, 446.5)	17 929	7.20 (6.39, 8.01)	318.9 (283.0, 354.7)	20 304	6.47 (5.75, 7.20)	117.8 (104.6, 131.1)	21 088	2.48 (2.21, 2.76)
2026	671.0 (598.1, 743.9)	8063	5.41 (4.82, 6.00)	413.5 (368.6, 458.4)	17 929	7.41 (6.61, 8.22)	334.7 (298.4, 371.1)	20 304	6.80 (6.06, 7.53)	124.0 (110.5, 137.4)	21 088	2.61 (2.33, 2.90)
2027	685.2 (613.1, 757.3)	8063	5.52 (4.94, 6.11)	425.3 (380.5, 470.0)	17 929	7.63 (6.82, 8.43)	350.3 (313.4, 387.1)	20 304	7.11 (6.36, 7.86)	130.1 (116.4, 143.8)	21 088	2.74 (2.45, 3.03)
2028	698.9 (627.6, 770.2)	8063	5.64 (5.06, 6.21)	436.9 (392.3, 481.4)	17 929	7.83 (7.03, 8.63)	365.4 (328.1, 402.7)	20 304	7.42 (6.66, 8.18)	136.3 (122.4, 150.2)	21 088	2.87 (2.58, 3.17)
2029	712.2 (641.6, 782.8)	8063	5.74 (5.17, 6.31)	448.2 (403.8, 492.6)	17 929	8.04 (7.24, 8.83)	380.1 (342.4, 417.8)	20 304	7.72 (6.95, 8.48)	142.4 (128.3, 156.6)	21 088	3.00 (2.71, 3.30)
2030	725.1 (655.3, 795.0)	8063	5.85 (5.28, 6.41)	459.3 (415.0, 503.5)	17 929	8.23 (7.44, 9.03)	394.4 (356.4, 432.4)	20 304	8.01 (7.24, 8.78)	148.6 (134.3, 162.9)	21 088	3.13 (2.83, 3.44)

	AlloSCT, first year			AlloSCT, second year			TFR			Total		
Year	Prevalent cases	YCPP (USD)	Prevalence costs (USD m)	Prevalent cases	YCPP (USD)	Prevalence costs (USD m)	Prevalent cases	YCPP (USD)	Prevalence costs (USD m)	Prevalent cases	Average YCPP (USD)	Prevalence costs (USD m)
2015	6.0 (5.0, 7.0)	159 432	0.96 (0.80, 1.12)	71.2 (59.9, 82.5)	9448	0.67 (0.57, 0.78)	84.3 (71.0, 97.7)	7558	0.64 (0.54, 0.74)	1205.4 (1014.3, 1396.5)	33 224	40.04 (33.70, 46.40)
2016	11.0 (9.3, 12.7)	161 312	1.77 (1.50, 2.05)	73.3 (62.1, 84.5)	9559	0.70 (0.59, 0.81)	89.6 (75.9, 103.3)	7529	0.67 (0.57, 0.78)	1264.2 (1071.2, 1457.3)	33 026	41.75 (35.37, 48.14)
2017	6.0 (5.1, 6.9)	163 525	0.98 (0.83, 1.13)	87.7 (74.7, 100.7)	9690	0.85 (0.72, 0.98)	96.8 (82.5, 111.1)	7442	0.72 (0.61, 0.83)	1323.6 (1127.8, 1519.5)	21 432	28.36 (24.16, 32.58)
2018	8.0 (6.9, 9.1)	165 354	1.32 (1.14, 1.50)	94.9 (81.3, 108.5)	9799	0.93 (0.80, 1.06)	103.3 (88.5, 118.1)	6994	0.72 (0.62, 0.83)	1383.6 (1185.2, 1581.9)	22 339	30.91 (26.50, 35.33)
2019	8.0 (6.9, 9.1)	177 076	1.42 (1.22, 1.61)	105.2 (90.5, 119.8)	10 493	1.10 (0.95, 1.26)	110.6 (95.2, 125.9)	7004	0.77 (0.67, 0.88)	1442.4 (1241.9, 1642.9)	19 413	27.99 (24.11, 31.89)
2020	8.0 (6.9, 9.1)	181 050	1.45 (1.25, 1.65)	113.8 (98.6, 129.1)	10 729	1.22 (1.06, 1.39)	116.0 (100.4, 131.6)	7008	0.81 (0.70, 0.92)	1497.4 (1296.4, 1698.4)	21 376	32.01 (27.71, 36.31)
2021	8.0 (7.0, 9.0)	183 544	1.47 (1.28, 1.65)	122.6 (106.7, 138.5)	10 877	1.33 (1.16, 1.51)	121.0 (105.3, 136.7)	6906	0.84 (0.73, 0.94)	1551.6 (1350.1, 1753.0)	18 228	28.28 (24.61, 31.94)
2022	8.0 (7.0, 9.0)	219 014	1.75 (1.53, 1.97)	132.1 (115.5, 148.6)	12 979	1.71 (1.50, 1.93)	125.9 (110.1, 141.6)	6468	0.81 (0.71, 0.92)	1610.1 (1408.1, 1812.0)	18 604	29.94 (26.19, 33.72)
2023	8.0 (7.0, 9.0)	209 838	1.68 (1.47, 1.89)	142.4 (125.2, 159.6)	12 435	1.77 (1.56, 1.98)	130.9 (115.0, 146.7)	6037	0.79 (0.69, 0.89)	1674.8 (1472.4, 1877.2)	15 937	26.70 (23.46, 29.93)
2024	8.0 (7.1, 8.9)	216 422	1.73 (1.54, 1.93)	153.2 (135.4, 171.1)	12 825	1.96 (1.74, 2.19)	135.8 (120.0, 151.6)	6054	0.82 (0.73, 0.92)	1741.6 (1538.8, 1944.4)	17 101	29.77 (26.33, 33.24)
2025	8.0 (7.1, 8.9)	211 175	1.69 (1.50, 1.88)	164.3 (145.9, 182.8)	12 514	2.06 (1.83, 2.29)	140.7 (124.8, 156.5)	6030	0.85 (0.75, 0.94)	1807.5 (1604.3, 2010.6)	14 408	26.04 (23.13, 28.97)
2026	8.0 (7.1, 8.9)	211 175	1.69 (1.50, 1.88)	175.6 (156.5, 194.7)	12 514	2.20 (1.96, 2.44)	145.5 (129.7, 161.3)	6030	0.88 (0.78, 0.97)	1872.3 (1668.9, 2075.7)	14 420	27.00 (24.06, 29.94)
2027	8.0 (7.2, 8.8)	211 175	1.69 (1.52, 1.86)	187.0 (167.3, 206.6)	12 514	2.34 (2.09, 2.59)	150.3 (134.4, 166.1)	6030	0.91 (0.81, 1.00)	1936.1 (1732.4, 2139.8)	14 431	27.94 (24.99, 30.88)
2028	8.0 (7.2, 8.8)	211 175	1.69 (1.52, 1.86)	198.4 (178.2, 218.7)	12 514	2.48 (2.23, 2.74)	154.9 (139.1, 170.7)	6030	0.93 (0.84, 1.03)	1998.7 (1794.8, 2202.6)	14 443	28.86 (25.92, 31.82)
2029	8.0 (7.2, 8.8)	211 175	1.69 (1.52, 1.86)	209.9 (189.1, 230.7)	12 514	2.63 (2.37, 2.89)	159.3 (143.5, 175.0)	6030	0.96 (0.87, 1.06)	2060.1 (1856.0, 2264.2)	14 453	29.78 (26.83, 32.73)
2030	8.0 (7.2, 8.8)	211 175	1.69 (1.52, 1.86)	221.5 (200.1, 242.8)	12 514	2.77 (2.50, 3.04)	163.5 (147.7, 179.2)	6030	0.99 (0.89, 1.08)	2120.3 (1916.1, 2324.6)	14 464	30.67 (27.70, 33.64)

Note: 95% confidence intervals are presented in parentheses.

Abbreviations: 1L, first-line; 2L, second-line; 3L+, third-line and later; AlloSCT, allogeneic stem cell transplantation; TKI, tyrosine kinase inhibitor; USD m, USD millions; YCPP, yearly cost per patient.

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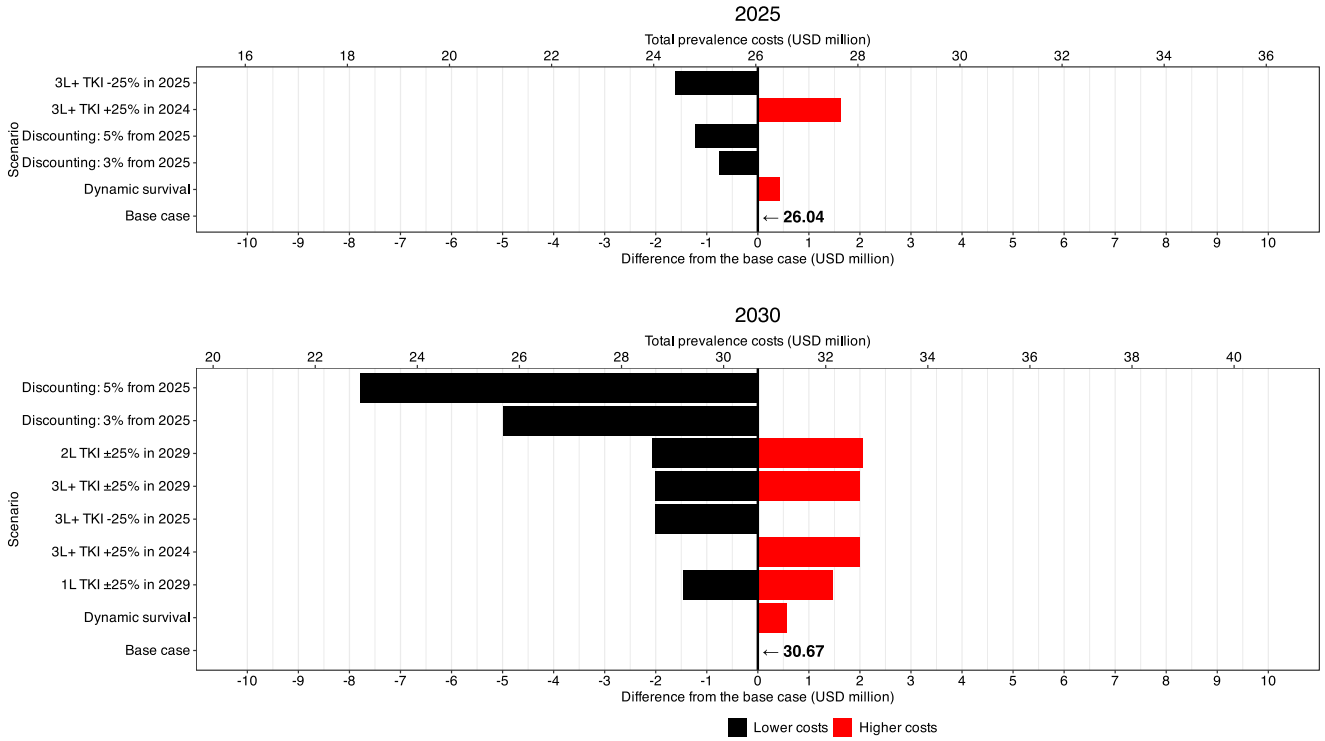


FIGURE 4 One-way sensitivity analysis of total prevalence costs of chronic myeloid leukaemia in Sweden for the years 2025 and 2030. Bars represent the difference from the base case (USD million) under various scenario assumptions. Black bars indicate lower costs, and red bars indicate higher costs relative to the base case.

Our findings broadly align with and extend previous studies. Gunnarsson et al.[2] predicted 15.1 CML cases per 100 000 persons in 2020 and 18.2 in 2030 under a fixed survival scenario, whereas our model estimated 14.60 and 20.35 cases for 2020 and 2030, respectively. These differences likely reflect our use of more recent follow-up data and different modelling approaches: Gunnarsson applied a period analysis and assumed constant excess mortality after 10 years, while we used empirical relative survival estimates without such constraints. Similarly, a recent French study[32] projected 18.74 prevalent cases per 100 000 persons in 2020 and 24.24 in 2030, which is in line with the rising trend observed in our estimates in Sweden.

Cost of illness studies, including ours, necessitate periodic updates as clinical practice and prices change. In our cost modelling, we incorporated disease states from a natural history model[23] informed by the Swedish CML register, with pricing data from Sweden's Prescribed Drug Register and TLV's Price and Decision Database.[19] We also treated TFR as a distinct disease state to capture its potential cost-saving impact. As clinical practice, treatment patterns and prices evolve, these estimates should be updated.

This study has several strengths. To our knowledge, it is the first nationwide, whole-population cost study for projecting CML prevalence costs using cancer register data, including both the Swedish Cancer Register and the Swedish CML register.[33] These results are specific to Sweden, although the methods are adaptable to countries with similarly structured population registers and comparable treatment patterns. We employed a previously developed natural history model for CML treatments[29] to predict the distribution of patients across disease states and integrated individual-level pricing data from the National Prescribed Drug Register and pharmaceutical and pricing register and database.[19] Projecting costs under two survival scenarios (fixed vs. dynamic survival) allowed us to quantify future uncertainty and showed that alternative survival assumptions had only a modest impact on the predicted number of prevalent cases and consequently the prevalence costs of CML in Sweden.

Nonetheless, several limitations warrant consideration. First, pricing assumptions for the years 2019–2025 relied on price ratios relative to 2018, and we assumed constant prices beyond 2025. Although discussions with experts in haematology suggested that substantial price shifts beyond 2025 are unlikely, this assumption might lead to a slight overestimation of costs in the later projection years, as historical trends indicated a decline in prices. Second, while the composition of TKI treatment was based on empirical data from 2007 to 2018 (Figure S6), we assumed that the distribution at each treatment line would remain constant beyond 2018. This may not fully capture future changes in treatment guidelines and prescribing patterns. Third, although adverse events, such as cardiovascular complications associated with nilotinib use,[34] and hospitalisation costs are part of direct health expenditures, these components were not included in our estimates. Notably, hospitalisation alone was reported to account for approximately 8.6% of the total yearly cost of care for CML patients,[35] suggesting that our estimates may modestly underestimate the true economic

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burden. Nonetheless, the one-way sensitivity analyses provided a structured approach to explore key sources of uncertainty in our projections. Fourth, our TFR data were limited to patients diagnosed between 2007 and 2012; extrapolation of these cohorts to later years may overestimate or underestimate the true number of patients eligible for TFR. Finally, cost estimates for patients in progression and alloSCT states were based on the aggregate data from Hernlund et al.[20] and adjusted yearly by Sweden's official price list,[21] rather than individual-level data from registers. Nevertheless, because relatively few patients transition to progression or alloSCT, the impact of these estimates on total prevalence costs is likely to be minimal.

Future policy initiative should (1) optimise prescribing to favour cost-effective TKIs when clinically appropriate, (2) expand eligibility for TFR when safety permits and (3) maintain post-marketing surveillance of newer generations of TKIs, such as asciminib, to confirm their long-term value for healthcare budget.

In conclusion, our study offers 15-year estimations and projections of total prevalence costs for CML that inform healthcare budget planning in Sweden, which are ready inputs for future economic evaluations and solid evidence for clinicians and policymakers aiming to manage CML care efficiently.

AUTHOR CONTRIBUTIONS

P.W.D. and T.D. acquired and cleaned the data. E.Y.-T.C. and T.D. cleaned the data. E.Y.-T.C. and F.D.M. conducted the formal statistical analyses. E.Y.-T.C., P.W.D., F.D.M., M.S.C., and S.H. developed the statistical methodology. E.Y.-T.C. and S.H. conceptualized the project and drafted the manuscript. E.Y.-T.C. prepared all figures and visualizations. E.Y.-T.C., P.W.D., F.D.M., T.D., L.S., M.B., M.S.C., and S.H. critically reviewed and revised the manuscript.

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CONFLICT OF INTEREST STATEMENT

Enoch Yi-Tung Chen, Paul W. Dickman, Fabrizio Di Mari, Takeda. Mark S. Clements and Shuang Hao: No relationship to disclose. Torsten Dahlén: (all outside of the submitted work) Consulting: Xspray pharma, Novartis. Leif Stenke: (all outside of the submitted work) Consulting: Xspray Pharma. Magnus Björkholm: (all outside of the submitted work) Grant committee: Incyte; Educational programme committee: Roche, Pfizer, Bristol Myers Squibb, Abbvie, Janssen, Takeda, Novartis; Consulting: WntResearch, Janssen-Cilag, and Schain Research; Research grant.

DATA AVAILABILITY STATEMENT

The statistical codes that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to ethical restrictions.

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